

Multimodal Simon Says

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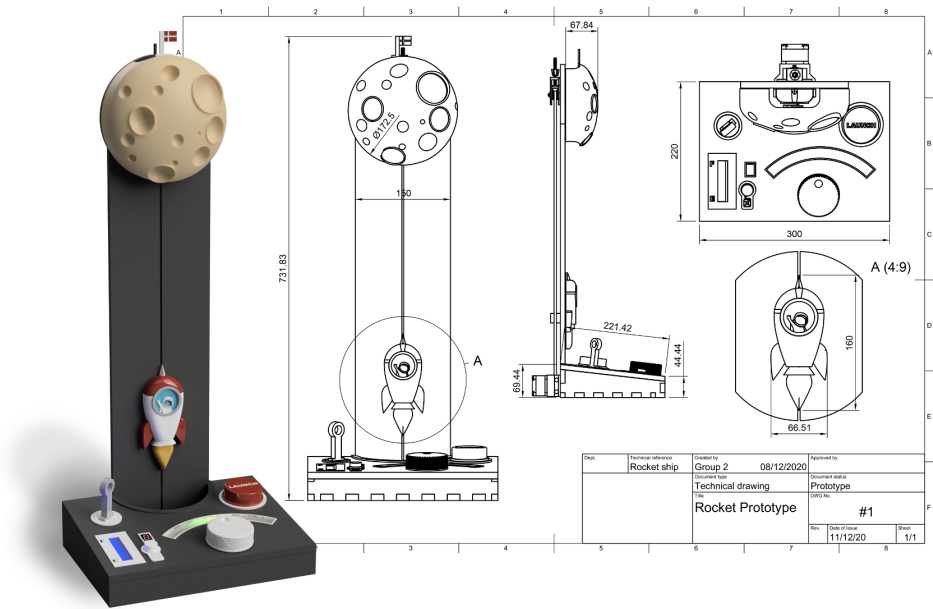


Fig. 1. XXXXXX

ABSTRACT

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1 INTRODUCTION

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2 PROTOTYPE DESIGN

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3 MAIN COMPONENTS

The prototype is split into two power rails. One of these is solely for the vibration motors, which operate properly in a relatively narrow voltage range. In addition to this, we also have a separate power source¹ for the motors, to avoid electrical interference, which could negatively impact the microcontrollers.

The other power rail is directly connected to another LiPo battery, since the components connected to this have a much wider operating voltage range. This means we avoid the efficiency

¹LiPo battery

loss from doing a voltage step-up. This means our power rails are, respectively, 3.4 volts² and 3.7 volts³.

3.1 Components

3.1.1 LiPo battery. As mentioned, we use two LiPo batteries each of which has a capacity of 820 mAh [7, p.1]. They have an operating voltage range between 2.75 and 4.2V [7, p.3], at respectively fully discharged and charged states. They are also very reliable and safe [7, p.4-6], which makes them well suited for a toy.

3.1.2 ATmega328P. The ATmega328P is used in our two agent boards, and control the actuators of the game. They also send information about button presses back to the controller upon request. We make use of eight of their I/O pins directly for the buttons and actuators, and we also make use of their I²C [6, p.173] capabilities to communicate with the controller board. These work in a wide voltage range and have a fairly low power draw.

Due to the simplicity of what we're using them for, and the reduced power draw, we've decided against running these at 16MHz using external crystals. This allows us to avoid stepping up the voltage, since running the chip at 8MHz is stable within the voltage range of a standard LiPo battery as seen in figure 2.

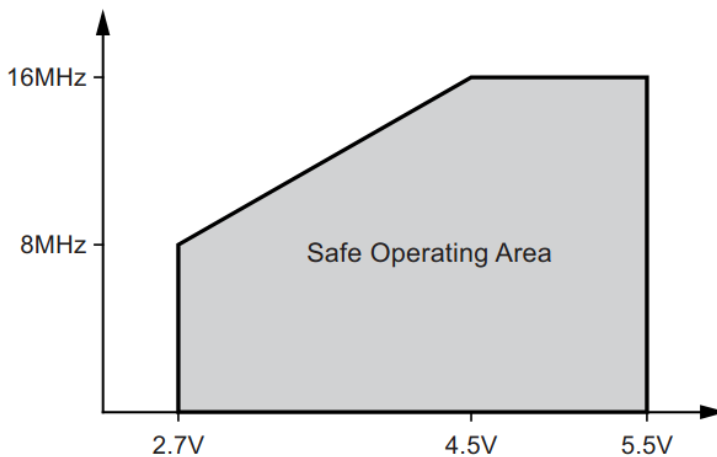


Fig. 2. Safe operating frequencies of the ATmega328P. Figure from [6, p.260].

To be more specific, we can support voltages between 2.7 and 5.5 volts [6, p.260], which is well within the range of the 3.7 volt LiPo battery. At the voltage we expect the microcontrollers to operate at for the majority of the time, their power draw should each be about 3.1mA according to the datasheet [6, p.268].

3.1.3 ESP 32. We use the ESP32-C3 super mini in our prototype as the controller of the game. The microcontroller requests information from the ATmega328P microcontrollers using I²C [10, p.4], and sends out appropriate commands. We use the hardware randomness module [10, p.5] within the ESP32, to generate sequences for the game. This ensures a reliable and unpredictable pattern. While the operating voltage of the ESP32 itself is only 3.0 to 3.6 [10, p.53], the module we're using

²See operating voltage of vibration motors.

³Actually a voltage range since it's connected directly to the battery.

has a power input pin that ranges from 3.3 to 6.0 volts [2, p.3]. Finding an exact expected power draw in the datasheet proved somewhat difficult due to the many possible configurations of the ESP32, but we have concluded it to be between 17 and 28mA [10, p.56].

3.1.4 Vibrating Motor. The vibrating motors are used for tactile instructions for the player. They operate at between 2.5 and 3.8 volts [5, p.5], which might lead one to think it could be run directly using the LiPo battery. However, the LiPo batteries actually reach up to 4.2 volts when fully charged, as mentioned in section 3.1.1, which would go over the limit of 3.8 volts. We therefore decided to step this down to 3.4 volts. The reason for this specific voltage will be explained in section 3.1.7. The motor has a maximum rated current draw of 75mA [5, p.5].

3.1.5 Bipolar junction transistor. To supply enough current to the vibrating motors, we can't just draw from a pin, since those are only rated for about 40mA [6, p.258]. We want more current than this, so we use the BC547 [4], which has a maximum collector current of 100mA [4, p.1]. This means our vibrating motors fit nicely within the current limit. The voltage rating is also more than sufficient, with both the collector-base and collector-emitter voltage rating being over 40 volts [4, p.1]. The emitter-base voltage rating is 6 volts [4, p.1], which is well above our 4.2 maximum voltage.

3.1.6 Light Emitting Diode. For the LEDs we simply used the ones at the lab, so we don't have the exact datasheet. This isn't strictly necessary though, as LEDs of this size are somewhat standard. We use both the standard single color LEDs and the multicolor LEDs. For the multicolor LEDs, we mostly found that they have a current rating of 20mA [1, p.1] per color. The forward voltage depends on the color, but for our usage, it should be respectively 2.3V and 3.4V for the red and green colors [1, p.1].

For the single color LED, we've used a standard datasheet we found on a reputable vendor site [9], and concluded the rated current to be, at most, 30mA [9, p.2], but we'll be a bit cautious and say 20mA, as that was the most common lower bound we could find for this type of LED. For the forward voltage it, again, depends on color, but we've chosen to assume a forward voltage of 2V⁴, as that will give a greater degree of caution than assuming a higher forward voltage. In practice, we've found the LEDs to be bright enough with these assumptions.

3.1.7 Low Dropout Regulator. To step down the voltage from the LiPo to 3.4 volts, we use a low dropout regulator [8]. These are a type of linear voltage regulator, with the advantage that they work even when the input voltage is close to the configured output voltage. This is exactly our use-case. Specifically, we're using the MIC29302WT, which has an output voltage range between 1.25V and 25V, and an input voltage range between 2.3V to 26V [8, p.4-7]⁵. The rated current is up to 3A, which is way more than needed for this project, but since the component was bought by a person in the group, they wanted the component to be very versatile for other future personal projects, which is also the reason for the adjustable output voltage. A more suited component for this prototype could've been something like the R1517J341B-T1-FE [3], which has a lower current rating of 500mA and outputs 3.4 volts.

The reason for choosing 3.4 volt over 3.3 volt, was to keep the vibration strength at a higher level for longer. While the LDO can maintain the voltage, it does still have dropout. According to the datasheet, for example, the output voltage might drop as low as 3V, when the battery has reached 3.5 volts, given a high enough current [8, p.12]. Though the current won't be that high in our case. To ensure proper vibration, we've therefore set the target voltage a bit higher than we actually

⁴This should be below the correct value.

⁵See note 7 in [8, p.7].

need, to hopefully ensure a better player experience for a wider voltage range.

The efficiency of the LDO is simply the output voltage divided by the input voltage, since it's a type of linear voltage regulator. This gives us an efficiency of about $\frac{3.4}{4.2} \approx 0.81$ in the worst case⁶.

3.2 Assembly

3.2.1 Box assembly.

3.2.2 Wiring.

3.3 Firmware

3.4 Electronics Calculations

4 LIMITATIONS AND FUTURE WORK

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5 CONCLUSION

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⁶The worst case efficiency is a fully charged battery.